

Programme	B. Sc. Computer Science				
Course Title	Soft Computing				
Type of Course	Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Fundamental Mathematics Concepts: Set, Functions, Logic 2. CSC2CJ101 – Fundamentals of Programming				
Course Summary	This course explores implementations of linked list and array-based data structures, delving into the inner workings of basic data structures including lists, stacks, queues, trees, and graphs.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the foundational principles of soft computing and the historical factors influencing its development.	U	C	Instructor-created exams / Quiz
CO2	Analyze the properties of Fuzzy sets and Fuzzy relations	Ap, U	P	Assignment/ Seminar
CO3	Apply fuzzy logic concepts to solve real-world problems, showcasing proficiency in designing and implementing fuzzy systems.	Ap, U	C	Seminar Presentation / Quiz
CO4	Master the concepts of Genetic algorithms and their operations	U	C	Practical Assignment / Seminar
CO5	Design and implement solutions using fuzzy logic, neural networks, and genetic algorithms for diverse applications.	Ap	P	Practical Assignment/ Seminar
CO6	Evaluate and present real-world scenarios where soft computing techniques can be effectively applied	Ap	P	Case study/ Project
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs
I	Introduction to Soft Computing (12 Marks)		7
	1	Overview of Soft computing, Hard Computing, and Hybrid Computing	2
	2	Areas and Applications of Soft Computing	1
	3	Basic Tools of Soft Computing- Fuzzy Logic, Neural Networks and Evolutionary computing	2
	4	Introduction to Fuzzy logic, Neural Networks, Genetic Algorithm, and Hybrid systems (Concepts only)	2
II	Introduction to Fuzzy Logic (20 Marks)		14
	6	Introduction to Fuzzy Logic	2
	7	Fuzzy sets and crisp sets	2
	8	Fuzzy relations and Crisp relations	2
	9	Tolerance and Equivalence Relations	2
	10	Fuzzy membership functions	3
III	11	Fuzzification and Defuzzification	3
	Advanced Fuzzy Logic (20 Marks)		14
	12	Fuzzy Rules and Fuzzy Reasoning	3
	13	Fuzzy Inference Systems- Mamdani and Sugeno models	4
	14	Fuzzy Control Systems	3
	15	Fuzzy Clustering (Concepts only)	2
IV	16	Fuzzy Neural Networks (Concepts only)	2
	Genetic Algorithm (18 Marks)		13
	17	Introduction to Genetic Algorithm	2
	18	Operators in genetic algorithm - coding - selection - cross over – mutation,	2
	19	Stopping condition for genetic algorithm flow.	2
	20	Constraints in Genetic Algorithm	2
V	21	Classification of Genetic Algorithm	3
	22	Genetic Programming (Concepts)	2
	Open Ended Module		12
		- Understand the different optimization techniques used. - Explore the real-life applications of soft computing techniques - Discuss hybrid soft computing techniques	

REFERENCES

1. S.N.Sivanandam and S.N.Deepa, "Principles of Soft Computing", Wiley India Pvt Ltd
2. D.K. Pratihar, "Soft Computing: Fundamentals and Applications", Alpha Science International Ltd

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	-	-	-	1	1						
CO 2	2	-	-	1	1	1						
CO 3	2	-	-	2	2	1						
CO 4	2	-	-	1	1	1						
CO 5	1	-	2	3	2	3						
CO 6	1	-	3	3	2	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5		✓		✓
CO 6			✓	